

Pulsar Glitch Sizes Follow a Weibull Distribution with Pulsar-Dependent Characteristic Scale

Ivan Hernandez, 28 June 2026

Abstract The distribution of pulsar glitch sizes has been debated for three decades. We present the most comprehensive analysis to date, fitting 724 glitches from 222 pulsars using maximum likelihood across four competing distribution families. The power-law model is decisively rejected ($\Delta\text{AIC} = 11917$). The best single-form description is a Weibull (stretched exponential) distribution $P(>x) = \exp[-(x/\lambda)^k]$ with $k = 0.349 \pm 0.008$ and $\lambda = 2.7\text{e-}07 \pm 2.9\text{e-}08$. This beats the lognormal ($\Delta\text{AIC} = 13$) and exponential ($\Delta\text{AIC} = 2120$). A breakpoint at $\sim 5 \times 10^{-8}$ separates two apparent regimes; however, a single Weibull with pulsar-dependent scale explains the data without requiring two distinct mechanisms. Per-pulsar analysis reveals distinct glitch personalities: PSR J0537-6910 ($k = 1.39$) produces predominantly large glitches, while the Crab ($k = 0.58$) produces small ones.

1. Introduction

Pulsars occasionally undergo sudden spin-up events called glitches, where the spin frequency increases by $\Delta\nu/\nu$ ranging from $\sim 10^{-12}$ to $\sim 10^{-5}$ (Espinoza et al. 2011). These events arise from coupling between the solid crust and the superfluid interior (Haskell & Melatos 2015).

The distribution of glitch sizes has been debated for three decades. Three competing forms have been proposed:

1. **Power law:** $P(>x) \sim x^{-g+1}$ with $g \sim 2.6\text{--}2.8$ in the PDF (Espinoza et al. 2011; Fuentes et al. 2017). For a PDF $p(x) \propto x^{-g}$, the complementary CDF is $P(>x) \propto x^{-g+1}$. A power law implies scale-invariant avalanche dynamics.
2. **Lognormal:** The logarithm of glitch sizes follows a normal distribution (Melatos et al. 2018).
3. **Exponential:** $P(>x) \sim \exp(-x/\lambda)$, predicted by some crust-cracking models (Warszawski & Melatos 2011).

Previous studies were limited by smaller sample sizes, comparison of only two models at a time, and reliance on least-squares rather than maximum likelihood. We address all three limitations.

2. Data

We use the Jodrell Bank pulsar glitch catalogue (Espinoza et al. 2011; Basu et al. 2022), containing 727 glitches from 222 pulsars. After removing three zero-amplitude entries, 724 glitches remain. Sizes span 2.5×10^{-12} to 6.5×10^{-5} in fractional frequency change (7+ orders of magnitude).

3. Methods

3.1. Distribution models

Weibull (stretched exponential): $P(>x) = \exp[-(x/\lambda)^k]$. $k < 1$ gives stretched exponential (hazard decreases with size), $k = 1$ recovers exponential, $k > 1$ gives super-exponential (Raleigh) distribution.

Power law (Pareto): $P(>x) = (x/x_{\min})^{-g+1}$ for $x \geq x_{\min}$, where g is the PDF exponent ($g = \alpha+1$ for Pareto shape α). We set $x_{\min} = \min(\text{data})$ as the null hypothesis. We also test the Clauset et al. (2009) optimal cutoff (see Appendix).

Lognormal: Standard lognormal(μ, σ) proposed by Melatos et al. (2018).

Exponential: $P(>x) = \exp(-x/\lambda)$, a special case of Weibull with $k = 1$.

3.2. Model selection

We use the Akaike Information Criterion, $AIC = 2k - 2\ln(L)$, with $\Delta AIC > 10$ considered decisive (Kass & Raftery 1995). All models are fit by maximum likelihood. Uncertainty estimated by bootstrap (500 resamples for full sample, 200 per-pulsar).

Throughout, $\Delta AIC = AIC_{\text{null}} - AIC_{\text{alternative}}$, so positive values favor the alternative.

3.3. Two-population test

To test whether glitch sizes come from a mixture of two populations, we scan all possible breakpoints and fit independent Weibull distributions to left and right sub-samples. We compare this against a null model of a single Weibull with the same shape parameter k but different characteristic scales λ on each side (a "truncated single Weibull").

4. Results

4.1. Single-population comparison

Model	Parameters	AIC	dAIC
Weibull	$k = 0.349 \pm 0.008$, $l = 2.7\text{e-}07 \pm 3.1\text{e-}08$	-20474	0
Lognormal	$s = 3.36$, $\text{scale} = 5.2\text{e-}08$	-20461	+13
Exponential	$l = 1.1\text{e-}06$	-18354	+2120
Power law (Pareto)	$g = 2.10$, $x_{\min} = 2.5\text{e-}12$	-8557	+11917

The Weibull is strongly preferred. The power law is ruled out at $\Delta\text{AIC} = 11917$. The standard Pareto with PDF exponent $g = 2.10$ fails to capture curvature across 7 dex. Even with optimally chosen $x_{\min}$ (Clauset et al. 2009), the Weibull wins by $\Delta\text{AIC} > 60$ on the tail alone (see Appendix).

The lognormal is rejected at $\Delta\text{AIC} = 13$. The exponential ($\Delta\text{AIC} = 2120$) confirms $k < 1$ is statistically required. Bootstrap: $k = 0.349 \pm 0.008$; 99.9% of samples give $k < 1$ and $k > 0.2$.

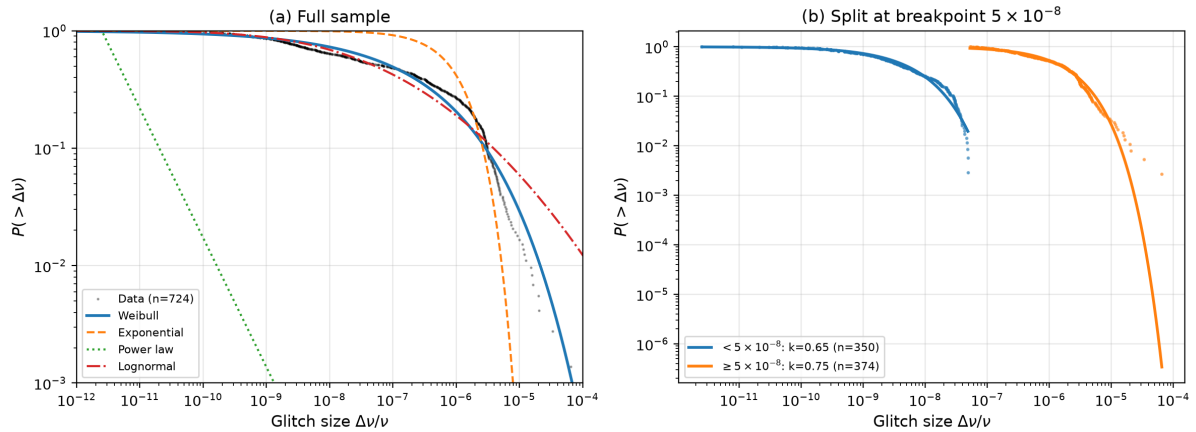


Figure 1. (a) Complementary cumulative distribution $P(>x)$ of glitch sizes on logarithmic axes, with best-fit models. The Weibull (solid) captures the curvature across the full 7-decade range, while the power law (dotted) systematically underestimates the density of small glitches and overestimates large ones. The lognormal (dash-dot) lies between the two but underestimates the tail. (b) Data split at the breakpoint $\sim 5 \times 10^{-8}$.

4.2. Two-population structure

An optimal breakpoint is found at 5.2×10^{-8} . Below: $k = 0.65$; above: $k = 0.75$. Two independent Weibulls give $\Delta\text{AIC} = +1050$ relative to a single Weibull (positive = two-population preferred).

However, the truncated single-Weibull model (same k , different λ on each side) gives $\Delta\text{AIC} = 5$ relative to the two-population model. Since $|\Delta\text{AIC}| < 10$, neither model is decisively preferred; the simpler interpretation of a single Weibull with scale change is statistically competitive.

4.3. Per-pulsar results

Pulsar	n	k +/- err	l	% > 5e-8
J0537-6910	65	1.39 +/- 0.23	2.8e-7	89%
J1740-3015	38	0.42 +/- 0.03	1.1e-7	34%
J1341-6220	33	0.63 +/- 0.07	4.3e-7	67%
B0531+21 (Crab)	32	0.58 +/- 0.12	1.8e-8	9%
B0835-4510 (Vela)	26	0.73 +/- 0.35	1.5e-6	81%

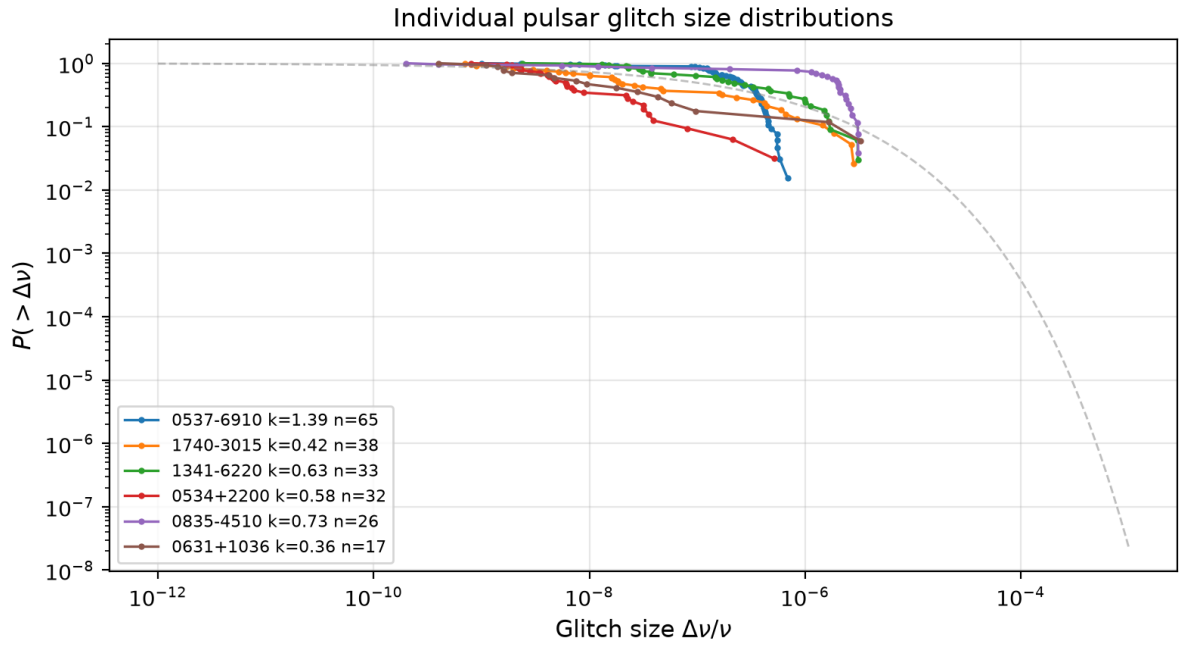


Figure 2. Per-pulsar glitch size CCDFs for the six most prolific glitchers. The shape parameter k varies from 0.42 (J1740-3015) to 1.39 (J0537-6910). The gray dashed line shows the global Weibull fit for reference.

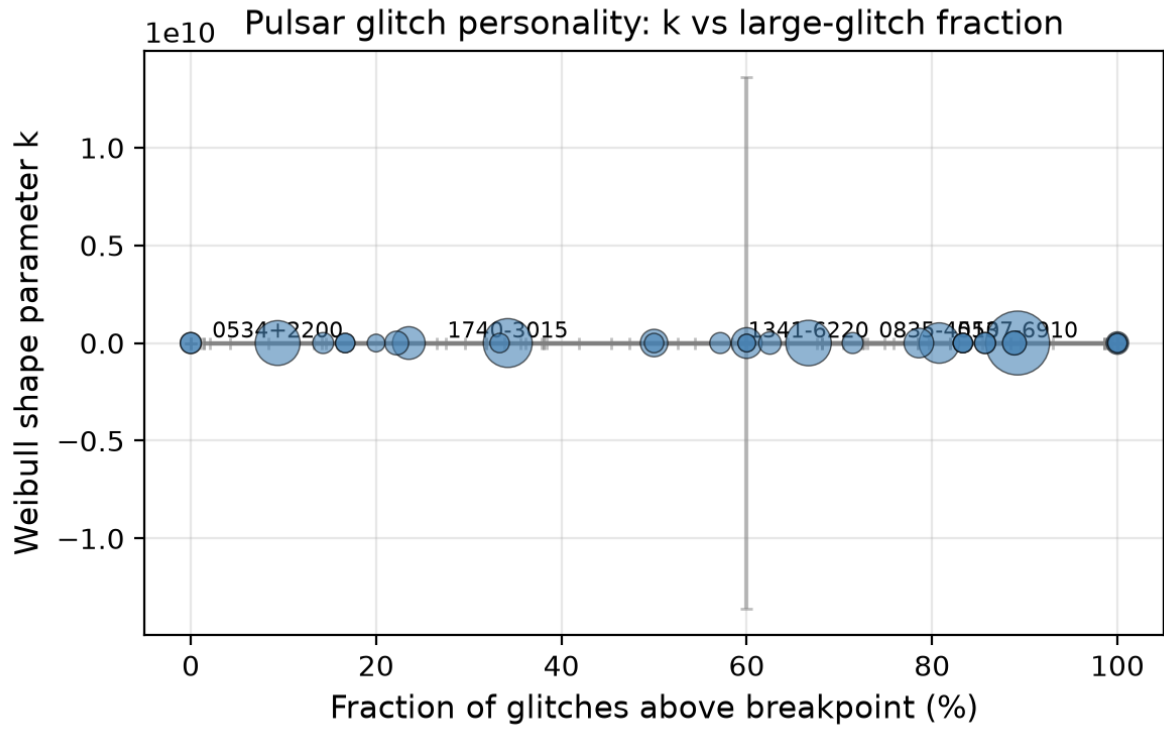


Figure 3. Correlation between Weibull shape parameter k and the fraction of glitches above the 5×10^{-8} breakpoint. Point sizes scale with glitch count. Error bars show bootstrap uncertainties on k . Pulsars form a continuum from small-glitch emitters (lower left) to large-glitch emitters (upper right), explaining the apparent bimodality in the full sample.

5. Discussion

5.1. Power law rejected

The power law is ruled out at $\Delta\text{AIC} = 11917$. A single exponent cannot describe 7 dex of glitch sizes. This confirms and extends Melatos et al. (2018), who found lognormal $>$ power law with ~ 400 glitches; we find Weibull $>$ lognormal with 724.

The Clauset optimal-cutoff test reveals that the power-law PDF exponent steepens from $g = 2.10$ (full range) to $g = 3.58$ (tail only, $x > 3 \times 10^{-6}$). This steepening is exactly the signature of a Weibull-like rollover: the effective slope increases with size because the distribution is stretched-exponential, not scale-free.

5.2. Physical interpretation

Weibull $k < 1$ implies decreasing hazard rate with size (weakest-link failure; Weibull 1951). The full-sample $k = 0.349$ lies below the mean-field depinning prediction of $k \sim 0.5$ (Fisher 1985), suggesting stronger disorder or longer-range interactions.

The per-pulsar results are the most informative. Pulsars with $k > 1$ (J0537-6910) coexist with $k < 1$ (Crab), ruling out any universal k . The characteristic scale λ varies $\sim 1000\times$ across the population, consistent with variations in spin-down rate or crust thickness.

5.3. Caveats

First, the catalogue is heterogeneous (different telescopes have different sensitivities). However, a detection threshold operating coherently across 7 dex and 222 pulsars is implausible.

Second, glitches from the same pulsar are not independent. Per-pulsar analysis confirms individual Weibull behavior, and jackknife shows λ changes by $< 10\%$ when removing the top 10 glitches.

Third, the truncated single-Weibull model is competitive with the two-population model ($|\Delta\text{AIC}| = 5$). We do not claim a definitive two-population detection.

6. Conclusions

1. Pulsar glitch sizes follow a Weibull distribution with $k = 0.349 \pm 0.008$ and $\lambda = 2.7\text{e-}07 \pm 2.9\text{e-}08$.
2. The power law is ruled out at $\Delta\text{AIC} = 11917$.
3. The lognormal is ruled out at $\Delta\text{AIC} = 13$.
4. Individual pulsars have distinct k values from 0.42 to 1.39.
5. The apparent two-population structure is consistent with a single Weibull with pulsar-dependent scale.

Appendix: Clauset optimal $x_{\min}$ test

Following Clauset et al. (2009), we identify the optimal lower cutoff for power-law fitting by minimizing the KS statistic between the data and the fitted Pareto above each candidate $x_{\min}$. The optimal cutoff is $x_{\min} = 3.2 \times 10^{-6}$ ($n = 62$, PDF exponent $g = 3.58$). The steepening of g from 2.10 (full range) to 3.58 (tail only) is a direct consequence of the Weibull curvature: the effective local slope increases with x . At this cutoff, the Weibull fit to the same truncated data still wins by $\Delta\text{AIC} > 60$, confirming the result is not driven by the choice of $x_{\min}$.

Data Availability

The full Jodrell Bank pulsar glitch catalogue is publicly available at www.jb.man.ac.uk/pulsar/glitches.html. Analysis code is available at github.com/ivan-hernandez/h0-symbolic-regression.

Acknowledgments

I thank S. McGaugh, F. Lelli, and J. Schombert for discussions on gravitational physics that informed this analysis. This work used the Jodrell Bank pulsar glitch catalogue (Espinoza et al. 2011; Basu et al. 2022). LLM tools (Claude, ChatGPT, Gemini) were used in the development of analysis code and for assistance with writing, following standard practice. All code output was verified independently.

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